EXERCISES IN 2D MOHR DIAGRAMS

Exercise 1

For each of the following states of stress (all stresses MPa) in which the *x*-direction points right and the *y*-direction points up (as usual when plotting a graph), solve for the principal stresses, σ_1 and σ_2, by drawing a Mohr diagram. In each case use the Mohr diagram to find the value of normal stress (σ_n) and shear stress (τ) on a plane oriented at 30° anticlockwise to σ_1.

(a) $\begin{bmatrix} 50 & 20 \\ 20 & 40 \end{bmatrix}$ (b) $\begin{bmatrix} 50 & 20 \\ 20 & -40 \end{bmatrix}$ (c) $\begin{bmatrix} 50 & -20 \\ -20 & 40 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & -40 \\ -40 & 80 \end{bmatrix}$

Method: (The first of these states of stress is shown as a worked example.)

$$\begin{bmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{bmatrix}$$

tensor notation

x-direction: $\sigma_n = \sigma_{xx}$; $\tau = \pm \sigma_{xy}$

y-direction: $\sigma_n = \sigma_{yy}$; $\tau = \pm \sigma_{yx}$

sign of shear stress needs to be determined

Mohr diagram convention

(a) Draw a square element and orient the stresses in the (x, y) coordinate frame. Note the tensor sign convention for the orientation of the stress components: look at the sides of the square that cut the negative coordinate axes; if the stress component is +ve it should point in the +ve coordinate direction on that side but if it is -ve it should point in the -ve coordinate direction.

(b) Determine the sign of the shear stresses in the Mohr circle convention (anticlockwise shear couple is +ve, clockwise shear couple is -ve).

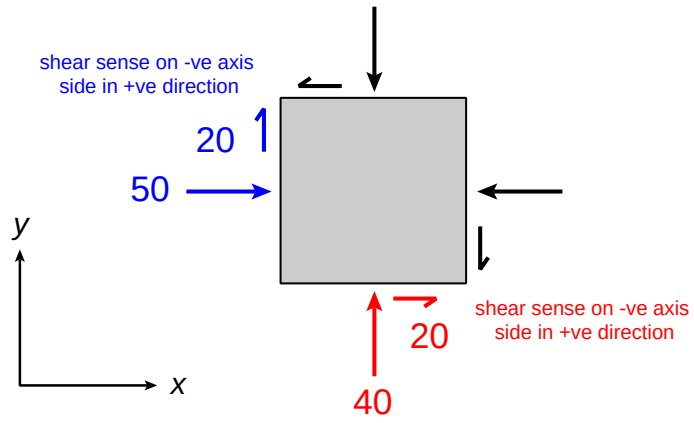
(c) Plot the two pairs of points $(\sigma_{xx}, \sigma_{xy})$ and $(\sigma_{yy}, \sigma_{yx})$ and draw the Mohr circle. Evaluate σ_1 and σ_2. Draw arrows in the correct orientation on your diagram of a square element that indicate the direction of σ_1 and σ_2.

(d) For the plane at 30° anticlockwise to σ_1 note that the normal is 90° - 30° = 60° clockwise to σ_1. Hence draw a line at 2 × 60° = 120° clockwise to σ_1 to find the point on the Mohr circle corresponding to this plane. Read off σ_n and τ. Draw a line in the correct orientation on your diagram of a square element that indicates the orientation of the plane, and add a lazy arrow to show the sense of shear.

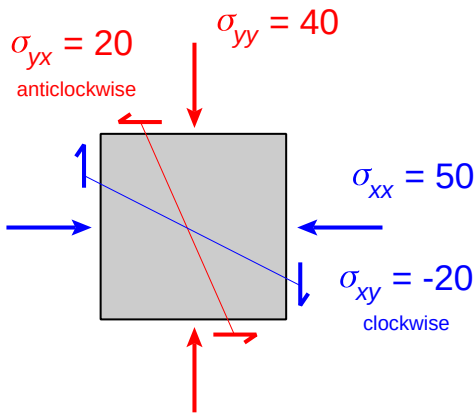
Worked solution for Exercise 1a

$$\begin{bmatrix} 50 & 20 \\ 20 & 40 \end{bmatrix}$$

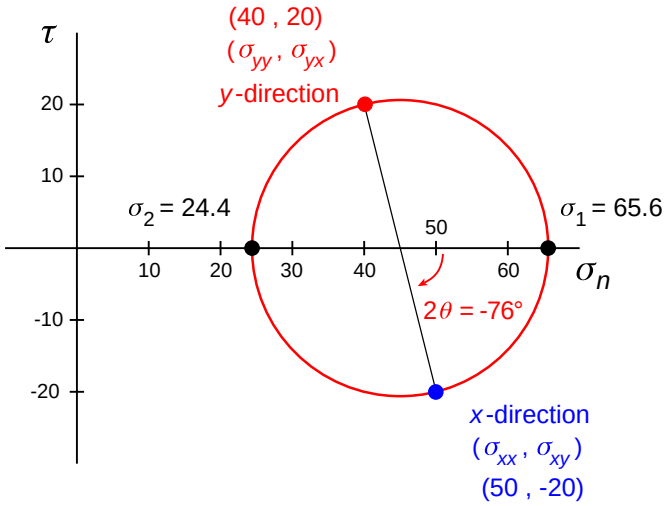
Step (a)

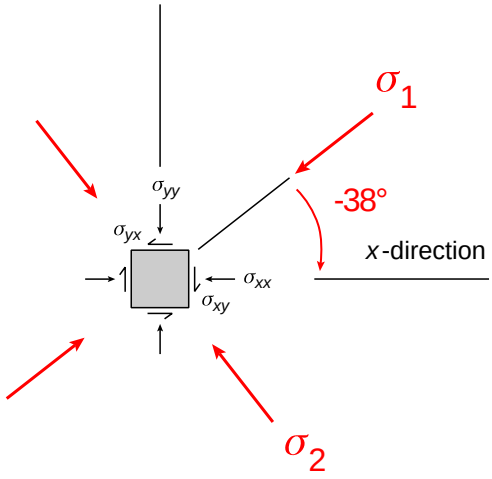
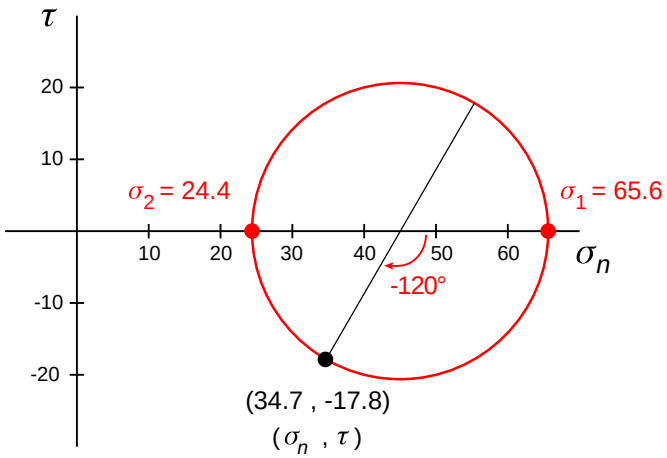
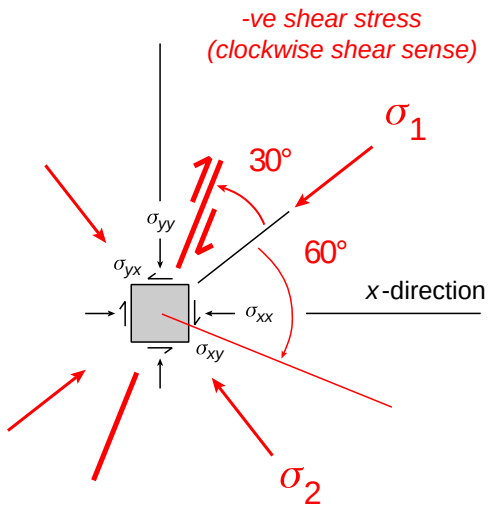


Step (b)



Step (c)





Step (d)